ELEC 221 – ELECTRIC CIRCUITS
COURSE SYLLABUS – FALL 2026

This is your course syllabus. Please download the file and keep it for future reference.

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory.
See: http://www.queensu.ca/encyclopedia/t/traditional-territories

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: https://www.queensu.ca/inclusive/.

TEACHING TEAM

COURSE INSTRUCTOR



Suzan Eren, PhD, PEng
Department of Electrical and Computer Engineering
E-mail: suzan.eren@queensu.ca
Office hours: Thursdays 10:30am-11:30am @WLH Rm 406

TEACHING ASSISTANTS

Graduate Teaching Assistants:	Labs, Quizzes, Tutorials, and OnQ
Ali Rezaei Aghoy (OnQ Manager)	Samira Fazlali Zadeh (Grading)
Prince Keelson (Tutorials)	Amyn Sorkhilaleloo (Grading)
Prosper Allo (Tutorials)	Matin Keshavarzi (Grading)
Ati Gavahian (Labs 1, 2, 3)	
Bahram Jahanbakhshi Pordanjani (Labs 4, 5, 6)	

Undergraduate Teaching Assistants:	Labs
Jack Buhr	Sanija Perera
Lucas da Silva	Michael Robles
Mar Garcia	Petra Vujasinovic

TUTORIAL TEACHING ASSISTANT

Name: Prince Keelson
Email: 24cbtt@queensu.ca
Office Hours: Wednesdays from 4:00pm-5:00pm @WLH 110 (In the basement. Knock on door to gain entry.)

Name: Prosper Allo
Email: 22xmg@queensu.ca
Office Hours: Mondays from 11:00am-12:00pm@WLH 405

Class Representative

Name: Nicolas Kaye
Email: 24fklv@queensu.ca

ELEC 221 (F 3-0.75-0.5 4.25)

Course Description

This course is an introduction to electrical circuits and circuit analysis techniques which are used in subsequent courses in electronics, power, and signals and systems. The course is augmented with online videos, problem solving and hands-on activities. These activities include learning technical skills such as building circuits, taking measurements, trouble-shooting, and transient analysis of both DC and AC circuits. To enhance the exposure to real-world applications, commercial software packages are used to design electrical circuits for preset applications, which are then built and analyzed.

Prerequisites: APSC 112 or APSC 114, APSC 171, APSC 172, APSC 174

Corequisites: MTHE 235 or MTHE 237 or MTHE 225 or MTHE 232 (Differential Equations)

Course Learning Outcomes (CLO)

By the end of this course, students should be able to:

CLO	Description	Indicator
CLO 01	Understand the basic circuit components and the fundamental laws of circuit theories (KCL, KVL, Ohm's law,...)	KB-ES (Circuits and Devices)
CLO 02	Derive the mathematical model of resistive, and first and second order circuits	KB-ES (Circuits and Devices)
CLO 03	Solve resistive circuits using techniques such as current voltage divider, mesh-current, node-voltage, Thevenin and Norton,	PA-Solve
CLO 04	Solve the initial condition and step responses of RC, RL and RLC circuits	PA-Solve
CLO 05	Solve sinusoidal steady-state response of RL, RC, and RLC circuits using techniques such as mesh-current, node-voltage, Thevenin and	PA-Solve
CLO 06	Calculate power consumption in RL, RC and RLC circuits under steady-state sinusoidal excitation	PA-Solve
CLO 07	Investigate the initial and step response of RL, RC and RLC circuit	IN-Analysis
CLO 08	Investigate the sinusoidal steady-state response of RL, RC and RLC circuits and power consummation is such circuits	IN-Analysis

COURSE EVALUATION

ASSESSMENT WEIGHTING

Activity	Date	Weight	CLO Alignment
Midterms (2)	**Date #1 TBD, Date #2 TBD** Please consult with OnQ announcements for latest information.	2@12.5% each = 25% total	1-5
Labs (6)	6 labs, alternate weeks. Please consult with OnQ announcements for latest information regarding lab schedule.	6 @ 5% each = 30% total	ALL CLOs
Final Exam (Proctored)	During the exam period.	45%	1-6
		100%	

ASSESSMENT DESCRIPTIONS

Midterms

There are two midterms in this course. These midterms are designed to provide learners with feedback on their knowledge. These midterms take place on the following dates: **TBD** and **TBD**. Details about the content of each midterm will be posted on OnQ closer to the date of the quiz.

Labs

There are six labs in this course. Please refer to the Laboratory Information and Schedule file posted on OnQ for further information. Here are a few things to note:

1. There are six lab sections (sections 005-010). You must check your SOLUS lab schedule to determine when and where to attend the labs. The Labour Day lab for Section 005 has been changed to Wednesday, September 16th from 6:30-9:30pm in BMH 314.
2. Lab files containing a description of the lab will be posted on OnQ.
3. You will be split into teams during the first lab, and this will be your team for duration of the term.
4. Labs will be assessed as follows:
    a. Pre-Lab Report (20%): Pre-labs must be finished individually before the lab takes place and will be checked for completion during the lab.
    b. Post-Lab Report (60%): One post-lab report is required per team. Post-lab reports should be submitted to OnQ by the due date, which is one week after your session.
    c. Attendance (20%): Attendance is taken at the start of each lab and is mandatory to receive credit for the lab. There is a 50% reduction in your attendance grade if you arrive more than 15 minutes late to your lab. Late by > 60 minutes: 0% attendance.

Final Exam

The final exam is closed book. Students must write their exam on the day and time scheduled by the University. You should not schedule vacations, travel, etc. during the exam period. The Term and Session Dates will indicate the final exam period session dates in each term.

GRADING

All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established Grade Point Index.

Accessing Your Final Grade

Your final grades will show on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

COURSE MATERIALS

Required Textbook

Electric Circuits (By: Nilsson and Riedel), 12th Edition, Pearson, 2023. The eText is available to purchase using the following link: https://campusebookstore.com/link/?id=98a9c8cf-fa8c-4008-a610-ed6897d3b33c&ref=7586b56c-5a1b-4be0-bd7a-8d6cda96cdd0&sid=ze8d1w7xqk1hr3vxg9jva57a.
We will cover Chapters 1-4 and 6-10, and Appendices A, B, F and H.

Other Material

All other course material is accessible via OnQ, including lecture slides, tutorials, announcements, lab instructions, and selected practice problems from the textbook.

Required Calculator

A Casio 991 is required. **ONLY** this type of non-programmable, non-communicating calculator will be allowed during tests and exams.

Suggested Time Commitment

This course represents a study period of one semester spanning 12 weeks. Learners can expect to invest on average 7-9 hours per week in this course. Learners who adhere to a pre-determined study schedule are more likely to successfully complete the course.

WEEKLY COURSE OUTCOMES

Week	Learning Outcomes
1	• Identify different applications of electric circuits in real life. • Define circuit variables (electric current, voltage, and power) from basic principles, and apply the passive sign conventions. • Identify ideal current and voltage (dependent and independent) sources, • Identify active and passive elements. • Recognize open and short circuits.
2	• Apply Kirchhoff's Laws: KVL, KCL. • Analyze electric circuits with dependent sources. • Manipulate complex electric circuits into simpler equivalent circuits: reduce resistive circuits, implement voltage and current dividers. • Identify different instruments used to measure current, voltage, and resistance, with a focus on both theoretical and practical considerations. • Apply Delta-Wye conversion technique in the time domain.
3	• Define new terminologies such as a planar circuit, node, essential node, path, branch, essential branch, loop, and mesh. • Use a systematic approach to solve electric circuits. • Apply node-voltage method to analyze electric circuits, considering independent and dependent sources and special cases. • Apply mesh-current method to analyze electric circuits, considering independent and dependent sources and special cases
4	• Determine (with reasoning) the most appropriate method to analyze a given problem (node-voltage vs. mesh-current). • Apply source transformation to simplify the process of circuit analysis, considering special cases. • Evaluate elements of Thevenin equivalent circuit for both independent and dependent sources. • Evaluate elements of Norton equivalent circuit for both independent and dependent sources.
5	• Solve simple circuits for maximum power transfer • Calculating the response of a linear system using superposition principle

Week	Learning Outcomes
6	• Solve simple electric circuits including inductors and capacitors to find voltages, currents, power and stored energy. • Evaluate equivalent inductance and capacitance for inductors and capacitors connected in parallel and in series (Capacitive/inductive circuit reduction).
7	• Determine the natural and step responses of first order RL circuits • Determine the natural and step responses of first order RC circuits
8	• Evaluate the general solution for step and natural responses of RL and RC circuits • Solve electric circuits including sequential switching • Evaluate the transient voltage responses of parallel RLC circuits and identify the damping of the circuit. • Evaluate the natural responses of parallel RLC circuits (overdamped, & underdamped)
9	• Evaluate the natural response of parallel RLC circuits (critically damped) • Evaluate the step response of parallel RLC circuits. • Evaluate the natural current responses of series RLC circuits and identify the damping of the circuits. • Evaluate the voltage step responses of series RLC circuits and identify the damping of the circuits.
10	• Define parameters of sinusoidal sources. Calculate Root-Mean-Square (RMS) of sinusoidal currents and voltages. • Determine the sinusoidal response of a circuit with linear passive circuit elements. • Use different representations of phasors (polar and rectangular), and use phasors inverse to find the original sinusoidal function, and the time-domain expression. • Calculate phase shifts of passive circuit elements. • Implement KVL and KCL in the frequency domain.

Week	Learning Outcomes
11	• Evaluate equivalent impedances and admittances in both series and parallel arrangements. • Apply Delta-Wye transformations and sources transformations in the frequency domain. • Apply Thevenin and Norton principles to find the equivalent AC circuits for both independent and dependent sources in the frequency domain. • Apply node-voltage / mesh-current to analyze a given AC circuit. • Calculating the response of a linear system using superposition principle in the frequency domain, for sources having the same frequency and different frequencies.
12	• Evaluate instantaneous, average and reactive power • Calculate the power for special circuits: purely resistive, purely inductive and purely capacitive. • Calculate the power factor for inductive loads (lagging) and capacitive loads (leading). • Calculate the rms (effective values) of voltages and currents in AC circuits.

COURSE COMMUNICATION

COURSE ANNOUNCEMENTS

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the **Begin Here** section of the onQ course site.

OFFICE HOURS

You will have the opportunity to interact with either the instructor or TAs through office hours. The instructor's office hours (also posted at the top) are Thursdays from 10:30-11:30am in WLH Rm 406. Please also see the first two pages of the syllabus for tutorial teaching assistant office hours.

ABSENCES (ACADEMIC CONSIDERATIONS) AND MISSED ASSIGNMENTS

For information on academic considerations due to extenuating circumstances, please review the information on the FEAS website. Note that unacceptable reasons include extra-curricular activities, travel plans, generally behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel is not an acceptable reason for granting academic considerations.

Confidential Matters

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document.

STANDARD QUEEN'S AND FEAS POLICIES

Netiquette

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

Following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.
5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

Student Code of Conduct

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation.
All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.
https://www.queensu.ca/nonacademicmisconduct/policies

Copyright

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the

instructor's ***express consent***. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

- Act at all times with devotion to the high ideals of personal honour and professional integrity.
- Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the Professional Engineers Ontario Code of Ethics. Information on policies concerning academic integrity is available in the Queen's University Code of Conduct, in the Senate Academic Integrity Policy Statement, on the Faculty of Engineering and Applied Science website, and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

INVALID EXAMS

An exam may be declared invalid in case of an interruption in an in-person examination; if the instructions in a remote or online exam were not followed; if the student uploads wrong materials; or if a situation arises where the integrity of the exam cannot be verified. If an exam is declared invalid, the student may be granted a re-write.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you including the Library, Student Academic Success Services (Learning Strategies and Writing Centre), and Career Services. Learners are encouraged to visit the Faculty of Engineering and Applied Science Current Students web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

Absences (Academic Considerations) and Academic Accommodations

For academic accommodations and considerations please review the information on the FEAS website.

Accommodations for Disabilities

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the QSAS website (https://www.queensu.ca/studentwellness/accessibility-services) to learn more about academic accommodations. To start the registration process with QSAS, click the ***Access Ventus*** button found on the Ventus student portal: https://www.queensu.ca/studentwellness/accessibility-services/ventus

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students: https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students

For questions or assistance with requesting Academic Consideration or Accommodation, contact the FEAS Program Advisor (Accommodations and Considerations) at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

Educational Technology	Accessibility Compliance Information
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/
MS-Teams	https://support.microsoft.com/en-us/office/accessibility-support-for-microsoft-teams-d12ee53f-d15f-445e-be8d-f0ba2c5ee68f
Zoom	https://zoom.us/accessibility

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

Religious Observance

Students in need of accommodation for religious observance are asked to speak to their professor within a week of receiving their syllabus. Note also that alternative assignments are considered a "reasonable accommodation" under the Ontario Human Rights Code. Students with questions about their rights and responsibilities regarding religious accommodation should contact the Chaplain Chaplain@queensu.ca.

OTHER HUMAN-RIGHTS BASED ACCESSIBILITY NEEDS

Students who have accessibility needs based on human-rights covered grounds, should inform their instructors within a week of receiving their syllabus. Student can also contact the contact the FEAS Program Advisor (Accommodations and Considerations) at engineering.aac@queensu.ca for guidance.

TECHNICAL SUPPORT

Some basic comfort level with basic hardware and software skills are required for this course. If you require technical assistance, please contact Technical Support.

SUPPORTIVE PERSONAL COUNSELLING

If at any time you find yourself feeling overwhelmed, anxious, sad, lonely, or distressed, consider confidential personal counselling and wellness services offered by the Faculty of Engineering and Applied Science and the Queen's student wellness services.